Multicore exercises

State fixed or growing work, worker count, overhead, and metric. Speedup compares elapsed times; efficiency divides speedup by workers.

CH07-Q01 Amdahl targets

A fixed workload has 90%parallel baseline time, 10%serial, and no overhead.

1. Find eight-worker speedup/efficiency.
2. Find the infinite-worker bound and distinguish finite attainability.
3. With 3%serial, find 128-worker speedup.
4. Compare eight workers with 5%serial against 16 workers with 10%serial.
5. For 100 workers and target 90 speedup, find minimum parallel fraction and serial percentage.

CH07-Q02 Fixed overhead and imbalance

10000 equal-cost units plus fixed launch 10 use 100 workers. One worker gets 2%of work, the other 99 evenly divide the remainder.

1. Find heavy/other worker work and total elapsed time.
2. Use 10010 baseline for speedup/efficiency.
3. Repeat for 5%heavy work.
4. What does work stealing improve, and which two overhead categories does it add?

CH07-Q03 Strong and weak scaling

1. A fixed problem takes 80 s on four cores versus 300 s on one. Find speedup/efficiency.
2. A four-times-larger problem keeps per-core work fixed and takes 90 s on four cores; the smaller one-core problem takes 75 s. Find weak efficiency. With linear single-core cost, find scaled speedup and state the compared sizes.
3. Ten additions remain serial; 100 or 10000 additions parallelize. At cost t per addition, find times/speedups for 10 and 100 workers.
4. Why does the small case improve 5.5→10 while the large reaches 91?
5. Does lower strong efficiency prove a race? Give at least three alternatives.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Multicore exercises (continued)

CH07-Q04 Address spaces and publication

1. U has one physical address space and uniform latency; M has private physical spaces per node. Classify them.
2. What correctness requirements apply to SMP object publication? Is coherence sufficient?
3. Explain when this acquire synchronizes and why r=42. Does acquiring 0 then reading data establish that relation?

```
data = 42
release store ready = 1
acquire load ready           // repeat until 1
r = data
```

4. In a separate incorrect variant, both data and ready are atomic and all four accesses use relaxed ordering. The writer stores data=42 then ready=1; the reader loads ready then data. Show an allowed ready=1, r=0 observation. Why is atomicity present without cross-object synchronization? If both objects are plain non-atomic with no other synchronization, why can results not be limited to 0 and 42?
5. Why does message passing require serialization/copy, routing, and optional ACK? Does ACK repair an application's unrelated race?

CH07-Q05 Hardware multithreading

A four-wide core often waits 200 cycles on an LLC miss while serving independent threads.

1. Compare fine/coarse-grained threading and SMT.
2. Does fine-grained threading guarantee lower single-thread latency?
3. Why can SMT fill issue slots but lower per-thread cache hit rates?
4. Average issue is 2.2 although single-thread ILP can fill four slots. Where may SMT help, and which downside metrics need measurement?

CH07-Q06 SIMD costs and Amdahl

Case A has 80 vectorizable one-cycle scalar operations and 20 nonvectorizable ten-cycle operations. Eight fully used lanes accelerate only the first group; ignore loads, setup, clock, and power changes.

Multicore exercises (continued)

1. Find each baseline/new time, accelerated baseline-time fraction, and overall speedup.
2. Independent case B has 100 one-cycle operations, 80 vectorizable. Find eight-lane speedup and effective-width-six, 75%-utilization result. Do not reuse A's time fraction.
3. How does vector length handle tails and varying lane counts compared with fixed-width SIMD?
4. Why need doubled SIMD width not double overall speedup? List excluded costs.

CH07-Q07 Network counts

Use a 4 x 4 endpoint grid. Mesh has adjacent local links; torus adds row/column wraps. Compare K 16 and a 16-port crossbar. Count each full-duplex physical link once; bisection is the minimum balanced-cut edge count.

1. Derive edge counts from rows/columns and count wraps.
2. Find mesh degree distribution, diameter, bisection, and the torus equivalents.
3. Give all eight wrap endpoint pairs and explain why removing any breaks the degree-four torus.
4. Find K 16 links, degree, diameter, bisection, and one-way cut bandwidth for per-link rate b.
5. Is crossbar endpoint degree the same? What grows quadratically?
6. Why does a uniform-traffic bisection summary not predict sparse-matrix performance?

CH07-Q08 Hierarchical Roofline

A multicore has PC=200 GFLOP/s and DRAM 40 GB/s, decimal units.

1. Derive the single-level roof and ridge.
2. Bound A at I=0.4 and B at I=12; identify bottlenecks.
3. Another phase performs $F = 10^9$ FLOPs at $P_C = 1000$ GFLOP/s. DRAM moves $Q_D = 0.25 \times 10^9$ bytes at $B_D = 100$ GB/s; shared memory moves $Q_S = 4 \times 10^9$ bytes at $B_S = 1000$ GB/s. Find intensities, memory bounds, and compute time.
4. Find cross-layer performance/time bounds. Explain DRAM(4, 400), shared(0.25, 250), and final(4, 250), including why 250 is not shared's plateau.
5. Is a measured 30 GFLOP/s sufficiently explained by being below the roof? Give at least two diagnostics.

Multicore exercises (continued)

References and attribution

David A. Patterson, John L. Hennessy. Computer Organization and Design: The Hardware/Software Interface, Revised Fourth Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study